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Portfolio: **https://github.com/Leo-Marcinkus/Engineering-Portfolio**

Seeking a Summer 2026 electrical engineering internship where I can contribute to the design and integration of electronic systems through hands-on experience in PCB design, embedded systems, circuit analysis, and measurement-driven troubleshooting.

Projects:

Golden Arduino Board: Designed and built a custom high-performance Arduino board. Created to streamline future work while completed under a tight deadline. Used Altium, interpreted and implemented multiple datasheets, and analyzed similar microcontrollers to design and assemble a board with multiple improvements.

Retro Glass Fuse Lamp: Created a retro aesthetic small-scale lamp utilizing glass fuse casings, LED Edison bulbs, and an integrated rechargeable battery circuit to expand my embedded systems and microelectronics knowledge. Improved upon the original inspiration by implementing ESP-NOW control, improvements in power efficiency, and good practices in battery safety resulting in a more robust design.

Alternative Radio Tuning: Designed an RF project involving a radio receiver utilizing an alternative capacitor-switching LC tuning strategy and a software-controlled frequency selection across the AM/FM broadcast bands. The objective is to find a more power-efficient and higher-performing RF tuning method than the modern method of digital synthesis. Implemented analog switching and amplification to receive real-time broadcasts and validate tuning using the NIST 60 kHz reference signal.

Apartment Rover: Used recycled parts to design a robot to autonomously navigate my apartment to broaden my knowledge of simple circuits. Utilized an Arduino with a proximity sensor and created the speed sensor, direction control, and speed regulation from scratch on a breadboard. Incorporated an OLED display for data visualization. Significantly improved debugging and prototyping skills with reuse constraints.

Sumo Robot: Developed a basic sumo robot to learn embedded systems. Primarily served to introduce embedded systems and MicroPython. Gained hands-on experience with motor drivers and IR/RF receivers.

Dynacone: Engineered a dynamically moving sports cone under a tight deadline to gain rapid prototyping experience. It is programmed to execute diverse training patterns and was engineered under a tight deadline with budget constraints. Utilized an Arduino, prototype PCB, motors, and a servo for precise movement.

Lab and Circuit Experience:

Advanced Electronic and Semiconductor Laboratory — August 2025 – December 2025

- Analyzed active and passive semiconductor devices (BJTs, diodes, photodiodes, MOSFETs) using oscilloscopes, DMMs, waveform generators, and various power supplies
- Performed I-V measurements, parameter extraction, and curve fitting to evaluate device performance and variability using industry-standard software
- Analyzed experimental results and correlated data with theoretical models in formal technical reports

PCB Design and Manufacturing — August 2025 – December 2025

- Designed multi-layer PCBs in Altium from schematic design to fabrication-ready layouts
- Utilized oscilloscopes to conduct signal integrity, crosstalk, and power-rail noise measurements
- Applied PDN and layout best practices through breadboard prototyping and PCB implementation

Embedded Software and Real-Time Operating Systems — August 2025 – April 2026

- Developed embedded firmware on ARM Cortex-M microcontrollers using C, registers, and memory-mapped peripherals
- Designed multi-threaded real-time applications using FreeRTOS with semaphores, mutexes, and task scheduling
- Implemented debugging, tracing, and profiling to validate timing and ensure optimal system performance

Electronics Design Lab — August 2024 – December 2024

- Designed, tested, and modeled DC motors, active op-amp low-pass filters, and encoder interface circuits for optimal signal conditioning
- Performed DC, transient, and frequency domain analysis in the SPICE-based software SIMetrix to optimize and debug circuit performance

Skills:

Hardware and PCB Design

- Analog circuit design in Virtuoso /Cadence/VLSI
- Design boards that pass validation quicker and with reduced debugging time
- Saturn PCB, multi-layer layouts, schematic testing, Altium, signal integrity, PDN design, power management

Electronics Analysis

- Perform circuit debugging and signal integrity analysis
- Analog circuit design, circuit testing, I-V characterization, LTSpice, ADS, HyperLynx

Embedded Systems

- Develop reliable firmware for rapid prototyping under firm deadlines
- STM32 boards, Arduino, microcontrollers, PWM, SPI/I²C/UART, Real-Time Systems, RISC-V

Programming Tools and Systems

- Debian and Arch-based Linux
- C/C++, Python/MicroPython, SystemVerilog, Assembly, CAD

Testing Tools and Verification

- Rapidly diagnose hardware problems and troubleshoot more effectively
- Oscilloscope, logic analyzers, DMM, data validation, waveform generators

Education:

University of Colorado Boulder, Boulder, CO

B.S. Electrical Engineering - College of Engineering and Applied Science – Expected May 2027

Accelerated M.S. Electrical Engineering – College of Engineering and Applied Science – Expected May 2028

Interests/Hobbies: Audio electronics and production, mechanical watches, guitar, piano, NBA, benchmarking components, Jimmy John's, EDC

Work Experience:

Lanzi Designs and Cabinetry – *Independent Contractor* — December 2022 – January 2026

- Measured, cut, and assembled cabinets with precision and professionally engaged with clients to ensure specifications
- Installed cabinetry, flooring, trim, and ceiling support for structural integrity and aesthetics

Eddie Bauer – *Sales Associate* — September 2022 – May 2023

- Operated the register, processed transactions, and assisted customers with purchases
- Resolved customer concerns and returns through clear communication and conflict management

Express – *Sales Associate* — May 2022 – August 2022

- Engaged with customers to provide styling advice and case-by-case recommendations
- Assisted in closing responsibilities to prepare the store for corporate reviews with executives